

■ Day 7 Assignment – Self Driving Car (Webots)

Objective

Implement your first autonomous vehicle controller in Webots and understand how software communicates with a simulated vehicle.

Install Required Software

- Webots R2025a
- Visual Studio Code
- Python 3.x
- NumPy
- OpenCV

Task 1 – Environment Setup

1. Open Webots.
2. Load the sample world: **vehicles** → **city.wbt**
3. Explore the Scene Tree and identify:
 - BMW X5
 - Camera
 - GPS
 - Gyroscope

Task 2 – Create Your Own Controller

- Create a new Python controller named **self_driving**.
- Replace the default controller of the BMW X5 with your own controller.

Task 3 – Robot API vs Driver API

Program 1 (Robot API)

- Print **Hello Webots!**
- Explain why the Robot API cannot directly drive the BMW X5.

Program 2 (Driver API)

- Initialize the Driver object.
- Print **Driver Initialized Successfully**.
- Explain why the Driver API is required for automobile models.

Task 4 – Autonomous Vehicle Motion

Without using the keyboard, write a Python program that automatically:

- Starts from rest.
- Moves forward.
- Reduces speed.
- Turns left.
- Continues moving for a few seconds.
- Stops automatically.

Rules

- No keyboard control.
- No manual intervention after pressing Play.
- Entire sequence must execute automatically.

Bonus Challenge

Modify your program so that speed and steering angle are stored in variables instead of hardcoded values.

Example:

MAX_SPEED = 20

TURN_SPEED = 8

LEFT_STEERING = 0.3

Submission (Google Drive)

Create a folder named **Day7** and upload the following structure:

Day7/

```
■■■ self_driving.py
■■■ screenshots/
    ■■■ scene_tree.png
    ■■■ hello_webots.png
    ■■■ driver_initialized.png
    ■■■ vehicle_motion.png
■■■ answers.pdf
```

answers.pdf

Answer the following questions:

1. What is the purpose of a controller in Webots?
2. What is the difference between the Robot API and the Driver API?
3. Why can't the Robot API directly control the BMW X5?
4. What is the purpose of the Driver API?
5. What is the name of the camera device?
6. What is the name of the GPS device?
7. What is the name of the Gyroscope device?
8. What happens if you remove the Driver controller from the vehicle?
9. Which functions did you use to control the vehicle's speed and steering?
10. What did you learn from today's task?

Note: If you get stuck, try reading the Webots documentation and official examples before asking for help. Research and debugging are part of robotics engineering.